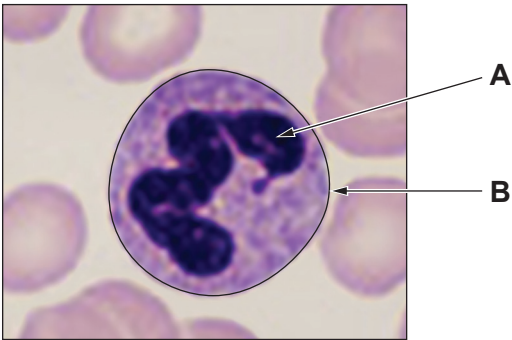


Examiner
only

2. (a) The photograph shows one labelled blood cell from the circulatory system.



(i) Complete the table by stating the name and function of structures **A** and **B**. [4]

Structure	Name	Function
A		
B		

(ii) The cell contains many mitochondria. State the function of mitochondria. [1]

.....

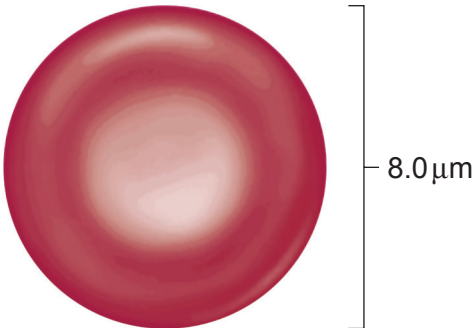
(b) Complete the table below by using words **from the list** to identify an **organ**, and a **cell** from the **human circulatory system**. [1]

heart lung palisade trachea phagocyte

Organ system	Organ	Tissue	Cell
circulatory		muscle	



(c) The diagram below shows a red blood cell.



(i) I. Complete the following calculation. [1]

diameter of red blood cell = 8.0 μm

radius (r) = 4.0 μm

Radius² = μm

II. Use your answer to part I. and the formula $3.14 \times \text{radius}^2$, to calculate the surface area of one side of a red blood cell to the **nearest whole number**. [1]

Surface area of one side of red blood cell = μm²

(ii) Red blood cells have a large surface area.
State the function of red blood cells and suggest how a large surface area helps red blood cells to be more efficient. [2]

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(iii) Red blood cells are specialised cells.
Give the meaning of the term *specialised cell*. [1]

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